

Cryptographic Security Posture Report

The-Ultimate-PCAP.pcapng · uploaded by user · analysed 2026-09-28 11:50:51.683204+00:00 · SHA-256 ecbca543fbe011a85c8876420fcb2c24...

POSTURE	RISK SCORE	SESSIONS	UPGRADES	FINDINGS	EVIDENCE
82/100	18/100	51	13	70	0.9
higher is better	CRITICAL	13 encrypted	STARTTLS / STLS	3 critical	1.0 = all visible

Finding severity

CRITICAL	Act now: credentials or content were exposed.	3
HIGH	Serious weakness: fix before the next mail flow that relies on it.	5
MEDIUM	Weak configuration: schedule a fix.	36
LOW	Minor or hygiene item.	0
INFO	Context, or evidence that was not visible.	26

Capture facts

Protocols	IMAP x7, POP3 x6, SMTP x38
TLS versions	TLS 1.2 x1, TLS 1.3 x5
Cipher suites	TLS_AES_128_GCM_SHA256 (0x1301) x2, TLS_AES_256_GCM_SHA384 (0x1302) x3, TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f) x1

Prioritised findings - showing the 40 most severe of 70

SEVERITY	FINDING	SESSION	EVIDENCE	RECOMMENDED ACTION
CRITICAL	Authentication pattern observed before encryption A USER/PASS/LOGIN/AUTH pattern was observed before a successful TLS upgrade.	tcp.stream.92	Plaintext authentication pattern observed Cleartext authentication command observed before TLS	Reject plaintext authentication and enforce TLS before credentials are accepted.
CRITICAL	Authentication pattern observed before encryption A USER/PASS/LOGIN/AUTH pattern was observed before a successful TLS upgrade.	tcp.stream.416	Plaintext authentication pattern observed Cleartext authentication command observed before TLS	Reject plaintext authentication and enforce TLS before credentials are accepted.
CRITICAL	Authentication pattern observed before encryption A USER/PASS/LOGIN/AUTH pattern was observed before a successful TLS upgrade.	tcp.stream.417	Plaintext authentication pattern observed Cleartext authentication command observed before TLS	Reject plaintext authentication and enforce TLS before credentials are accepted.
HIGH	Certificate hostname does not match SNI The certificate identity did not match the observed server name.	tcp.stream.93	SNI: dovecot.weberlab.de Certificate CN: jvw-vm08-int-dns	Install a certificate containing the correct mail host in its SAN extension.

SEVERITY	FINDING	SESSION	EVIDENCE	RECOMMENDED ACTION
HIGH	Certificate is expired The captured leaf certificate is outside its validity period.	tcp.stream.93	Valid until: 2026-08-14 09:33:19+00:00	Replace the certificate and monitor renewal before expiry.
HIGH	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.92	Protocol: IMAP Server port: 143	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
HIGH	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.416	Protocol: POP3 Server port: 110	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
HIGH	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.417	Protocol: POP3 Server port: 110	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Certificate is self-signed The leaf certificate is self-signed and may not be trusted by clients.	tcp.stream.93	Subject and issuer: CN=jw-vm08-int-dns	Use a certificate issued by the organisation's approved CA or a trusted public CA.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.7	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.13	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.14	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.15	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.16	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.17	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.18	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.

SEVERITY	FINDING	SESSION	EVIDENCE	RECOMMENDED ACTION
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.19	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.20	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.21	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.22	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.23	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.24	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.25	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.26	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.27	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.28	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.29	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.30	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.

SEVERITY	FINDING	SESSION	EVIDENCE	RECOMMENDED ACTION
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.31	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.32	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.33	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.34	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.35	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.36	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.37	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.38	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.39	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.40	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.41	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
MEDIUM	Email session is not protected by TLS Application-layer email traffic was observed without a visible TLS session.	tcp.stream.42	Protocol: SMTP Server port: 25	Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.

Showing the 40 most severe of 70 findings in this document. The HTML and JSON reports contain every finding, and the dashboard lists them per session.

Sessions (51)

SESSION	PROTOCOL	TLS VERSION	CIPHER SUITE	UPGRADE	RULES	ML	SCORE
tcp.stream.93	SMTP 25	TLS 1.2	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)	upgraded	HIGH	HIGH	78
tcp.stream.92	IMAP 143	UNKNOWN	UNKNOWN	cleartext	CRITICAL	CRITICAL	75
tcp.stream.416	POP3 110	UNKNOWN	UNKNOWN	cleartext	CRITICAL	CRITICAL	75
tcp.stream.417	POP3 110	UNKNOWN	UNKNOWN	cleartext	CRITICAL	CRITICAL	75
tcp.stream.7	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.13	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.14	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.15	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.16	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.17	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.18	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.19	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.20	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.21	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.22	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.23	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.24	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.25	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.26	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.27	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.28	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.29	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.30	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.31	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.32	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.33	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.34	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.35	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.36	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18

SESSION	PROTOCOL	TLS VERSION	CIPHER SUITE	UPGRADE	RULES	ML	SCORE
tcp.stream.37	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.38	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.39	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.40	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.41	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.42	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.43	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.44	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.45	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.46	SMTP 25	UNKNOWN	UNKNOWN	cleartext	MEDIUM	MEDIUM	18
tcp.stream.418	POP3 110	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.419	POP3 110	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.420	POP3 995	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.421	POP3 995	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.422	SMTP 587	TLS 1.3	TLS_AES_128_GCM_SHA256 (0x1301)	upgraded	LOW	LOW	0
tcp.stream.423	SMTP 465	TLS 1.3	TLS_AES_128_GCM_SHA256 (0x1301)	upgraded	LOW	LOW	0
tcp.stream.424	IMAP 143	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.425	IMAP 143	TLS 1.3	TLS_AES_256_GCM_SHA384 (0x1302)	upgraded	LOW	LOW	0
tcp.stream.426	IMAP 143	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.427	IMAP 143	TLS 1.3	TLS_AES_256_GCM_SHA384 (0x1302)	upgraded	LOW	LOW	0
tcp.stream.428	IMAP 993	UNKNOWN	UNKNOWN	upgraded	LOW	LOW	0
tcp.stream.429	IMAP 993	TLS 1.3	TLS_AES_256_GCM_SHA384 (0x1302)	upgraded	LOW	LOW	0

Recommended actions

- 1
- Require TLS before authentication and prefer implicit TLS or successful STARTTLS/STLS.
- 2
- Reject plaintext authentication and enforce TLS before credentials are accepted.
- 3
- Replace the certificate and monitor renewal before expiry.
- 4
- Use a certificate issued by the organisation's approved CA or a trusted public CA.

5	Install a certificate containing the correct mail host in its SAN extension.
6	Capture the complete ClientHello and ServerHello for a definitive assessment.
7	Capture the complete handshake, including the key-share or key-exchange fields.
8	Capture a complete full handshake before concluding certificate security.
9	Enforce TLS 1.2 or TLS 1.3 and remove SSLv3/TLS 1.0/TLS 1.1.
10	Prefer ECDHE/DHE or TLS 1.3 ephemeral key exchange for forward secrecy.
11	Prefer AES-GCM or ChaCha20-Poly1305 and remove obsolete cipher suites.
12	Require valid, unexpired certificates with a complete issuer chain and matching SAN.
13	Require authentication only after a successful TLS upgrade.

What this report does and does not claim

It reports what the capture shows. Values labelled as negotiated come from the server's own handshake; values the client merely offered are labelled as offers. Missing evidence is reported as UNKNOWN or PARTIAL and never as a weakness: an unobserved certificate, a certificate encrypted inside TLS 1.3, or a truncated handshake are absences of evidence, not findings. The analysis is passive. No email content is read, stored or decrypted, and an incomplete capture cannot prove an attack - it can only show that something is worth checking. Evidence completeness for this capture: 0.9 (1.0 means every field the analyzer looks for was visible).